

Day 4: Network Cable Types and Tools Theory : Types of cables (Ethernet, fiber optics) and tools. Practical : Create a straight-through Ethernet cable:-

Tasks :-

1.List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

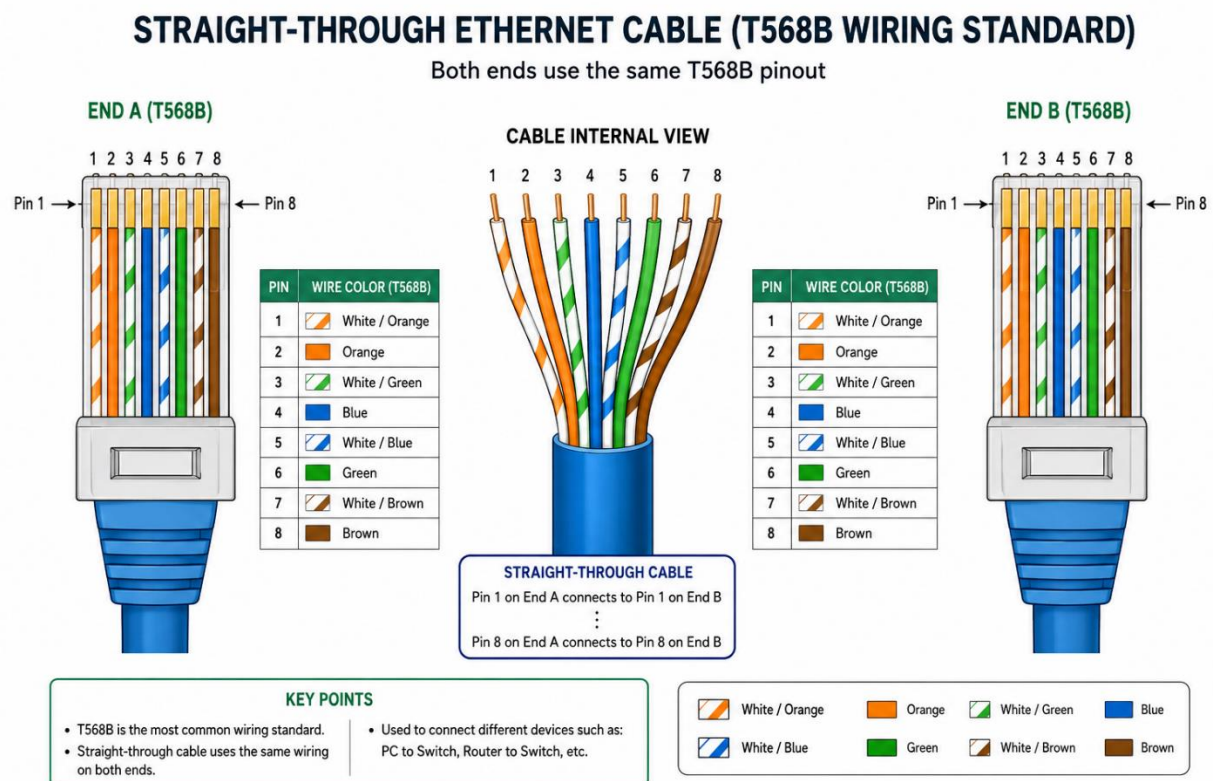
Ans. :-

Feature	Ethernet Cable (Copper)	Fiber Optic Cable
1. Transmission Method	Transfers data using electrical signals through copper wires.	Transfers data using light pulses through glass or plastic fibers.
2. Material	Made of twisted-pair copper wires (Cat5e, Cat6, Cat6a, Cat7, Cat8).	Made of very thin strands of glass or plastic fiber.
3. Speed	Common speeds are 100 Mbps, 1 Gbps, 2.5 Gbps, 5 Gbps, and 10 Gbps depending on the cable category.	Can support 10 Gbps, 40 Gbps, 100 Gbps, 400 Gbps, and even higher in enterprise networks.
4. Maximum Distance	Works best up to 100 meters without a repeater.	Can transmit data for several kilometers (single-mode fiber can exceed 40 km or more).
5. Signal Loss	Signal weakens over longer distances.	Very low signal loss, making it ideal for long-distance communication.
6.Electromagnetic Interference (EMI)	Can be affected by electrical equipment, motors, or power cables.	Completely immune to electromagnetic interference.
7. Installation	Easier to install, terminate, and repair.	Requires specialized tools and trained technicians for installation and splicing.
8. Cost	Lower cable and installation cost.	Higher initial cost, but better for high-speed, long-distance networks.

9. Durability	More resistant to bending and rough handling.	More fragile and must be handled carefully.
10. Security	Easier to tap or intercept electrical signals.	More secure because tapping the cable is difficult and usually detectable.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

Ans.:-



3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Ans. :-

Step-by-Step Guide: Attaching an RJ45 Connector to a CAT6 Cable (T568B Standard) :-

Tools Required :-

- CAT6 UTP cable
- RJ45 connector (CAT6 compatible)
- RJ45 crimping tool
- Wire stripper
- Cable cutter
- Network cable tester (recommended)

Step 1: Measure and Cut the Cable:-

Measure the required cable length and cut the CAT6 cable cleanly using the cable cutter.

Step 2: Strip the Outer Jacket:-

Using the wire stripper on the crimping tool, remove about 2–3 cm (1 inch) of the outer insulation. Be careful not to damage the inner wires.

Step 3: Untwist the Wire Pairs:-

Separate the four twisted pairs and untwist only as much as necessary. Keeping the twists close to the connector helps maintain signal quality.

Step 4: Arrange the Wires in T568B Order:-

Straighten the eight wires and arrange them from Pin 1 to Pin 8 in this order:

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

Step 5: Trim the Wires Evenly:-

Hold the wires firmly and trim their ends so they are all the same length, leaving about 12–13 mm ($\frac{1}{2}$ inch) exposed.

Step 6: Insert the Wires into the RJ45 Connector:-

Hold the RJ45 connector with the gold contacts facing upward and the locking clip facing downward. Carefully slide all eight wires into the connector until each wire reaches the front and the cable jacket enters the connector to provide strain relief.

Step 7: Crimp the Connector:-

Insert the RJ45 connector into the crimping tool and squeeze the handles firmly until the tool completes the crimp. This pushes the metal contacts into the wires and secures the cable jacket inside the connector.

Step 8: Repeat for the Other End:-

For a straight-through Ethernet cable, use the same T568B wiring order on the second end.

Step 9: Test the Cable :-

Use a network cable tester to verify that all eight pins are connected correctly (Pin 1 → Pin 1, Pin 2 → Pin 2, ..., Pin 8 → Pin 8). If any wire is incorrect or not connected, cut off the connector and crimp a new one.

**4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.**

Ans.:-

If I were setting up a gaming café with 10 PCs for LAN gaming, I would choose CAT6 Ethernet cables and an RJ45 crimper tool.

I would use CAT6 Ethernet cables because all 10 computers would be located in the same room or within a short distance of each other. Ethernet cables are ideal for distances up to 100 meters and provide high-speed data transfer (1 Gbps to 10 Gbps, depending on cable quality and distance). They also offer low latency, which is very important for multiplayer LAN games where fast response times and stable connections are required. In addition, CAT6 cables are much more affordable than fiber optic cables, making them the best choice for a small gaming café with a limited budget.

To attach RJ45 connectors to the CAT6 cables, I would use an RJ45 crimper tool. This tool securely fixes the connectors onto the cable, ensuring a reliable network connection. After crimping, each PC would be connected to a network switch, and the switch would then be connected to the router to provide both LAN connectivity and internet access.

I would not choose fiber optic cables or a fiber cleaver because fiber is mainly designed for long-distance communication, very high bandwidth, and large networks such as data centers, college campus backbones, and Internet Service Providers (ISPs). Fiber installation is more expensive and requires specialized tools and technical skills. For a gaming café with only 10 PCs in one location, the additional cost and complexity are unnecessary.